Operating Systems

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Meet the instructor

- Lecturer at CS Department, Faculty of Computers and Information.
- Bachelor in Computer Science at Assiut University.
- Master's in Computer Science & Engineering at UConn.
- Ph.D in Computer Science at Brown University.
- ex-Apple, Microsoft, Cadence, Y Combinator.
- Ubuntu open-source community.

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Grading

School year work

- Project 1: **15** pts
- Project 2: **15** pts

Exams

- Midterm : **20** pts – 25 Questions MCQ
- Final: **50** pts – 60 Questions MCQ

No grades on attendance, but:

- Every lab session you attend = 1 pt
- You can apply 5 points to each project.

Lab Sessions

- Come prepared with questions.
- TAs are here to help.
- What you do in the lab session builds up for your project 1 and 2.

References

- Stanford CS111
- Operating Systems: Principles and Practice (2nd Edition)
  https://github.com/CS162Textbook/Operating-Systems-Principles-and-Practice-2nd
- Lectures are recorded
- Resources are shared on Microsoft Teams (including memes)

Code of Conduct

- **Respect** each other. We are all learning.
- **Engagement** is maximum learning.
- **Teamwork** is lovely. **Cheating** is bad.
- **No questions?** More content.
- **Use AI** – as long as it helps you understand.
- You will not understand everything. **Do it anyway!**

Any questions?
No stupid question!

If something comes up later, contact abdelrahman@aun.edu.eg

What is an Operating System?

An operating system (“OS”) is software that allows people to run programs on a computer.

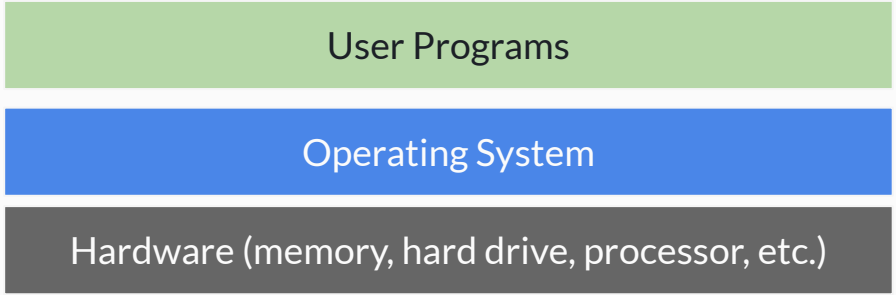
Examples: iOS, Android, Windows, macOS, Linux

You may think mostly of the user interface of the operating system, but an operating system does so much more!

What is an Operating System?

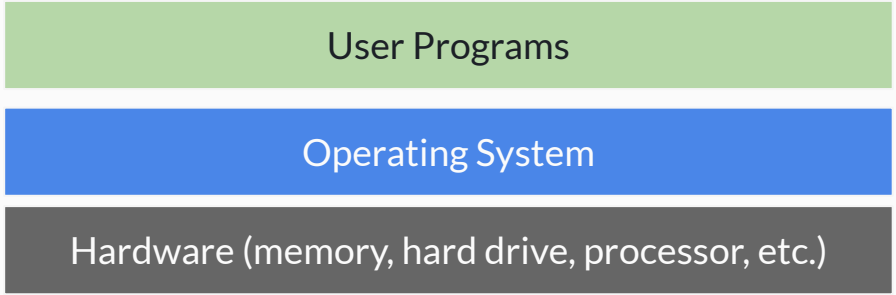
The operating system sits between the hardware and user programs. It manages shared resources and provides functionality for programs to run. It manages things like:

- Processor (CPU): decides what program gets to do work and for how long
- Memory (RAM): decides what programs get to use what areas of memory
- Hard Drive: decides how the disk is used to store files



Key Operating System Responsibilities

- **Concurrency:** manages different tasks running simultaneously
- **Memory (RAM):** allows system memory be shared among several tasks
- **Files:** allows many files, for many different users, to share space on disk
- **I/O devices:** allows many devices to operate concurrently and be shared
- **Networks:** allows groups of computers to work together
- **Security:** allows interactions while protecting participants from each other



What is an Operating System?

So far, when you've written programs, you don't think about any of this.

The OS abstracts away complexity from programs.

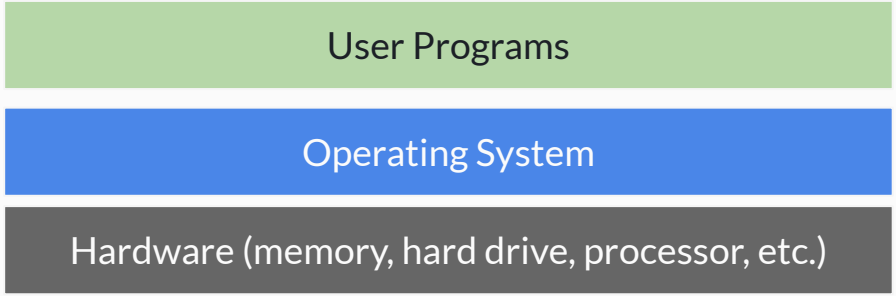
- Don't have to coordinate with other programs for who gets to use what memory.
- Don't have to coordinate with other programs for who gets to run when.

We can directly leverage operating system functionality in our programs. System calls are functions the operating system provides that we can call in our code. For example, in Linux:

- the open() function lets us open a file on disk
- the fork() function lets us spawn a new program instance (!)

This class

- Learn what functionality is exposed by operating systems to make the programs that we write more powerful.
- Learn how the operating system provides that functionality and how it acts as an interface to the computer hardware.



This class

1. **Filesystems** - How can we design file systems to manage files on disk, and what are the tradeoffs inherent in designing them? How can we interact with the filesystem in our programs?
2. **Processes** - How can our program create and interact with other programs? How does the operating system manage processes?
3. **Threads** - How can we have concurrency within a single process? How does the operating system manage concurrency?
4. **Virtual Memory** - How can one set of memory be shared among several processes? How can the operating system manage access to a limited amount of system memory?
5. **Miscellaneous** - Various terms you will encounter during your career in software engineering.
6. **The Shell** - building your own!

Next

How can we design file systems to manage files on disk, and what are the tradeoffs inherent in designing them? How can we interact with the filesystem in our programs?

Any questions?
No stupid question!

Let's discuss any doubts or concerns.
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